TimeSiam: A Pre-Training Framework for Siamese Time-Series Modeling

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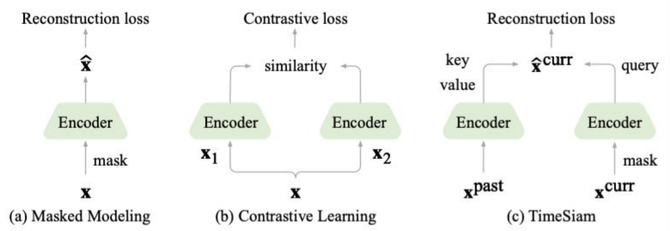
Poster

Emphasize Temporal Correlation Modeling based on Siamese Networks



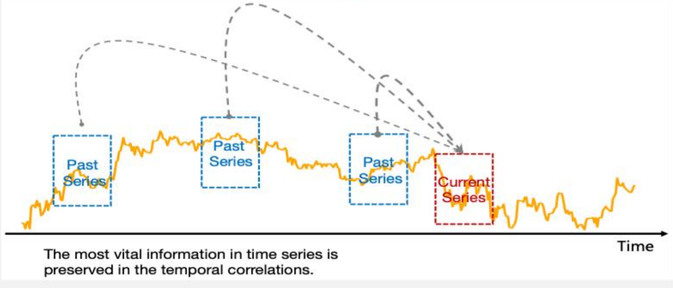
Compare Previous work:

① **Masking**: Reconstruct the masked series

② **Contrastive**: Repulse different series (negative pairs), while attracting two augmentations from the same series (positive pairs)

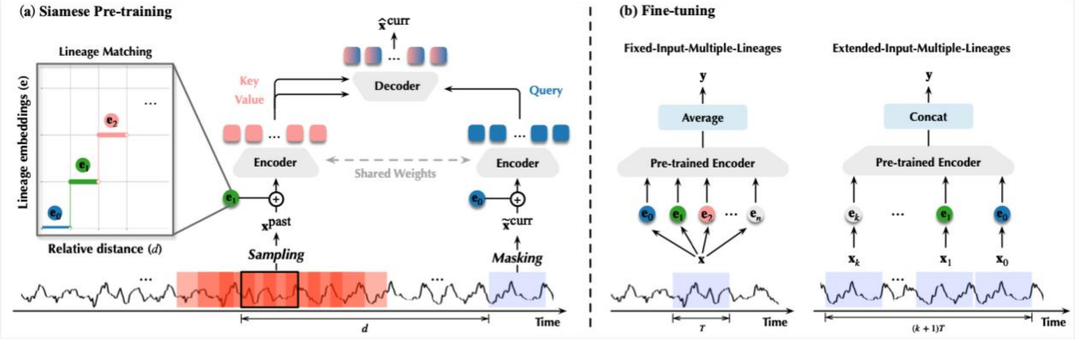
③ **TimeSiam**: Reconstruct masked current series from randomly sampled past series

Key Insight: Optimizing Encoders to accurately Model Temporal Correlation



All the code, data and scripts are in https://github.com/thuml/TimeSiam

TimeSiam: A Simple yet Effective Self-supervised Pre-training Framework



➢ **Siamese Pre-Training:** Learning Correlations Between Siamese Subseries

① Siamese Sampling

② Lineage Matching

③ Representation Learning & Aggregation

④ Series Reconstruction

Masked current serves as the query, past acts as both the key and value

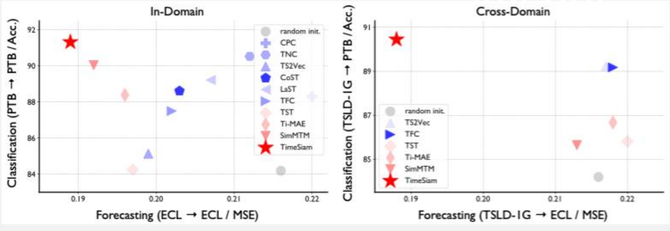
$$\begin{aligned}\widehat{\mathbf{h}}_d &= \mathrm{LayerNorm}\left(\widetilde{\mathbf{h}}_e^{\mathrm{curr}} + \text{Cross-Attn}\left(\widetilde{\mathbf{h}}_e^{\mathrm{curr}}, \mathbf{h}_e^{\mathrm{past}}, \mathbf{h}_e^{\mathrm{past}}\right)\right)\\ \mathbf{h}'_d &= \mathrm{LayerNorm}\left(\widehat{\mathbf{h}}_d + \text{Self-Attn}\left(\widehat{\mathbf{h}}_d, \widehat{\mathbf{h}}_d, \widehat{\mathbf{h}}_d\right)\right)\\ \mathbf{h}_d &= \mathrm{LayerNorm}\left(\mathbf{h}'_d + \mathrm{FFN}\left(\mathbf{h}'_d\right)\right).\end{aligned} \tag{4}$$

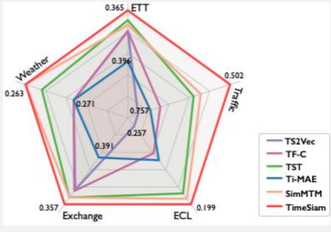
➢ **Fine-tuning:** Applying Pretrained Siamese Networks and Diverse Lineage Embedding to Enhance Fixed or Extended Representations

Outperforms advanced per-training baselines covering 13 Standard Benchmarks

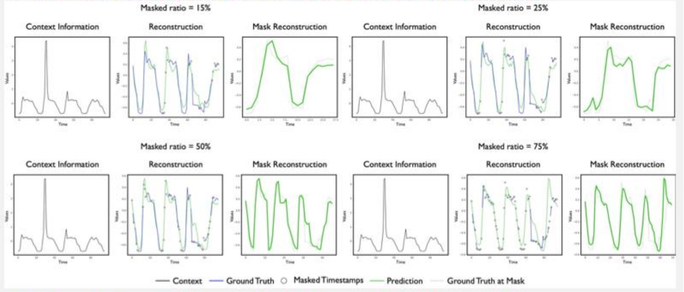
➢ **Two typical time series analysis tasks:** Forecasting and Classification

➢ **Compared to** 11 advanced **Masked Modeling** and **Contrastive Learning** Method under in- and cross domain, single and mixed datasets setting

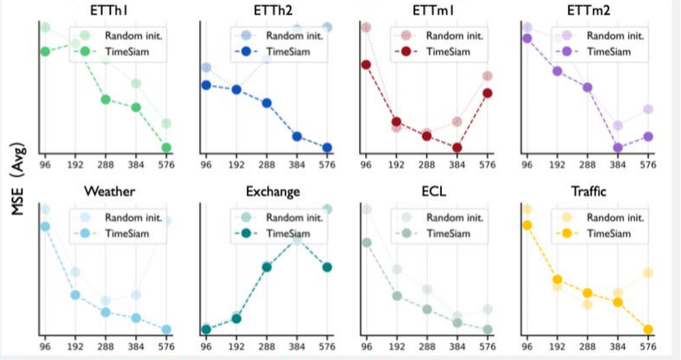




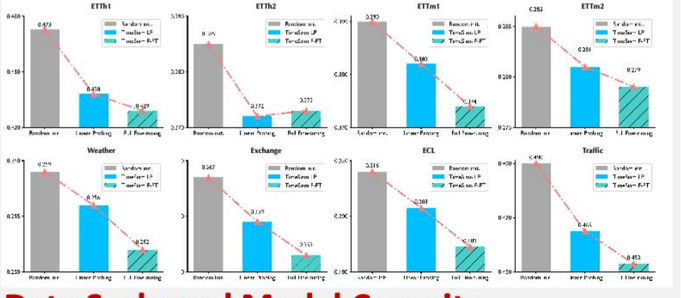
Different Masked Ratio Showcases



Adapt to Extended-Length Input



Linear Probing and Fine-tuning



Data Scale and Model Capacity

Pre-train	Traffic	ECL
Random initiation	0.490	0.216
TimeSiam in-domain	0.453	0.189
$\text{TimeSiam}_{\text{Base}}$ $\text{TSLD}_{500\text{M}}$	0.462 (+5.7)	0.189 (+12.5)
$\text{TimeSiam}_{\text{Base}}$ $\text{TSLD}_{1\text{G}}$	0.454 (+7.4)	0.188 (+12.5)
$\text{TimeSiam}_{\text{Large}}$ $\text{TSLD}_{1\text{G}}$	0.433 (+11.6)	0.185 (+14.4)

the positive correlation between **data-model size** and **the final performance**